The Cone Generated by Positive Semidefinite Mean Polynomials

Date: October 03, 2025

Mehdi Ghasemi, Salma Kuhlmann

Introduction

We noticed that certain classes of (homogeneous) polynomials, such as Positive Semidefinite (PSD) Diagonal minus Tail (D-T) forms [GM01], and more generally, those derived from the Arithmetic-Geometric Mean inequality, like Sums Of Nonnegative Circuit (SONC) polynomials, can be identified with nonnegative mean polynomials.
In this paper we study psd forms that are weighted power means, denoted by $M_{q,p}(Y,w)$, where $Y=(Y_1,\dots,Y_m)$ an m-tuple of symbols, $w=(w_1,\dots,w_m)\in\mathbb{R}^m_+$. In particular we investigate the cone of nonnegative mean real n-variate forms of degree $2d$, denoted $M_{n,2d}(X,\alpha)$, where $X=(X_1,\dots,X_n)$, and $\alpha\in\mathbb{N}_0^n$, and its relation to other important cones :

- $\mathcal{P}_{n,2d}$: The cone of all Positive Semidefinite (PSD) forms in m variables of degree $2d$. Our cone is a subset: $\mathcal{M}_{n,2d}\subseteq\mathcal{P}_{n,2d}$.
- $\Sigma_{n,2d}$: The cone of Sums of Squares (SOS) polynomials.
- $\mathcal{C}_{n,2d}$: The cone of Sums of Nonnegative Circuit (SONC) polynomials.
  A key object of study is the sum of the SOS and SONC cones, known as the **SOSONC cone**, defined by M. Schick:

  $$SOSONC=(\Sigma+\mathcal{C})_{n,2d}$$

  In **Section 1** (Preliminaries on mean polynomials), we will introduce the preliminary concepts of weighted power means, denoted $M_{q,p}(Y,w)$, $q,p\in\mathbb{N}_0$, $Y=(Y_1,\dots,Y_m)$ an m-tuple of symbols, $w=(w_1,\dots,w_m)\in\mathbb{R}^m_+$ and the fundamental tool of Jensen's inequality and prove (see Proposition 1.1) that if $p<q$, then $M_{q,p}(Y,w)$ is PSD. We identify certain classes of nonnegative polynomials as instances of $M_{q,p}(Y,w)$ for appropriate choices of q,p,Y, and w.
  In **Section 2** (Nonnegative Circuit Polynomials): We re-prove [GM01, Cor2.2] and [DIdW01, Thm2.4] using mean polynomials, giving necessary and sufficient conditions for nonnegativity of D-T forms, respectively circuit polynomials (refer to the numbers in section 2).
  In **Section 3** (The cones $M_{2d,0}(\underline{X},\alpha)$ and $M_{1,0}(Y,\lambda)$): We prove **Proposition 3.1**, the forms $M_{2d,0}(X,\alpha)$ are psd D-T for any $d\in\mathbb{N}$, and any $\alpha\in\mathbb{N}_0^n$ with $|\alpha|=2d$. The cone generated by the forms for is the cone of sums of PSD D-T forms. That is, the forms $M_{2d,0}(X,\alpha)$ are

PSD for any $d \in \mathbb{N}$.
In **Theorem 3.1**, we prove that the SONC cone, $\mathcal{C}_{n,2d}$ is the cone generated by the forms $M_{1,0}(Y, \lambda)$.
In **Section 4**, we study $M_{2d,p}(X, \alpha)$ with $|\alpha| = 2d$, $p|2d$ in connection to SOBS and SOS.
We prove in Proposition 4.1 that $M_{2d,0}(X, \alpha)$ is SOBS and psd D-T, $|\alpha| = 2d$. In Theorem 4.1 we continue investigating $M_{2d,p}(X, \alpha)$ with $d = 1, 2, 3$, $p|2d$ and prove that they are SOS for $\forall n \in \mathbb{N}$ (Reduction to Hilbert case, and induction on number of variables).
In **Section 5** (General Case: Beyond the SOSONC cone), the ultimate goal is to move towards a comprehensive understanding of the general form $M_{q,p}(Y, w)$.
In particular we examine the Choi-Lam form Q and the second Robinson form $\hat{R}$ (see section 5.1).
In [MSc01] it is observed that Q is SONC [MSc01, page 31], but is not SOSONC [MSc01, Lemma 4.2.5, page 107]. It was further observed by Reznick [Rez07] that $\hat{R}$ can be obtained from Q by a linear change of variables. In section 5.2, we establish moreover, that

$$Q(x, y, z, w) = 4M_{1,0}((x^4, y^4, z^4, w^4), (1, 1, 1, 1))$$

and accordingly

$$\hat{R}(x, y, z, w) = 8M_{1,0}(G, (1, 1, 1, 1))$$

where $G = \frac{1}{2}(x + w - y - z, y + w - x - z, z + w - x - y, w - (x + y + z))$. Mehdi to check the computation.
Therefore $\hat{R}$ is a separating form for the cone of mean polynomials and the SOSONC cone.
In **Section 6** (Positivestellensatz and relation the the cone of $2d$ powers).

In **Section 7** (Application to Polynomial Optimization): Mehdi: to check the algorithmic aspect of means.

Section 1: Preliminaries on mean polynomials

Proposition 1.1 Define $\varphi(x) = x^{\frac{2q}{p}}$, where $q > p$. The second derivative of φ is calculated below:

$$\varphi''(x) = (\frac{q}{p})(\frac{q}{p} - 1)x^{\frac{q}{p}-2} = \frac{q}{p}(\frac{q}{p} - 1)x^{\frac{2q}{p}-2}$$

Since $q > p$, the coefficient $\frac{q}{p}(\frac{q}{p} - 1)$ is always positive and if q is even then φ'' is always positive and hence the function φ is convex. By Jensen' inequality, we have

$$\varphi(\frac{\sum w_i x_i}{\sum w_i}) \leq (\frac{\sum w_i \varphi(x_i)}{\sum w_i}),$$

where $w_i \in \mathbb{R}_+$. Replacing x_i with x_i^p, we get

$$\varphi(\frac{\sum w_i x_i^p}{\sum w_i}) \leq \frac{\sum w_i \varphi(x_i^p)}{\sum w_i},$$

and hence

$$(\frac{\sum w_i x_i^p}{\sum w_i})^{\frac{q}{p}} \leq \frac{\sum w_i x_i^q}{\sum w_i},$$

or equivalently $M_p(x,w) \leq M_q(x,w)$. Raising both side to the power $c = \mathrm{lcm}(q,p)$, we conclude that the function

$$\mathcal{M}_{q,p} = M_q^c(x,w) - M_p^c(x,w),$$

is always nonnegative polynomial functions in $x = (x_1, \ldots, x_n)$.

Basic properties of M_p:

- $\lim_{p\to-\infty} M_p(\mathbf{x}, w) = \min(x_1, \ldots, x_n)$.
- $\lim_{p\to\infty} M_p(\mathbf{x}, w) = \max(x_1, \ldots, x_n)$.
- $\lim_{p\to 0} M_p(\mathbf{x}, w) = \left(\prod_{i=1}^n x_i^{w_i}\right)^{1/\sum_{i=1}^n w_i}$.
- If $p < q$, then $M_p(\mathbf{x}, w) \leq M_q(\mathbf{x}, w)$. Equality happens when $x_1 = x_2 = \cdots = x_n$.

Section 2: Nonnegative Diagonal minus Tail and Circuit polynomials

Theorem 2.1 Let the function f(x) be defined as:

$$f(x) = \sum_{i=1}^n \beta_i x_i^{2d} - \mu x^\alpha,$$

where $x = (x_1, \ldots, x_n)$ with $x_i \geq 0$, $\beta_i > 0$, and $\alpha = (\alpha_1, \ldots, \alpha_n)$ with $\alpha_i > 0$. The monomial x^α is defined as $x^\alpha = \prod_{i=1}^n x_i^{\alpha_i}$. We assume the degree is conserved, i.e., $\sum_{i=1}^n \alpha_i = 2d$.
The function $f(x)$ is positive semidefinite (PSD), meaning $f(x) \geq 0$ for all $x \in \mathbb{R}_+^n$, if and only if the following condition holds:

$$\mu^{2d} \prod_{i=1}^n \alpha_i^{\alpha_i} \leq (2d)^{2d} \prod_{i=1}^n \beta_i^{\alpha_i} \quad (*).$$

Proof:
The proof consists of two parts for the "if and only if" statement.
Part 1: ($\Leftarrow$)
Assume the inequality $(*)$ holds. We want to show that $f(x) \geq 0$. This is equivalent to showing:

$$\sum_{i=1}^n \beta_i x_i^{2d} \geq \mu x^\alpha \quad \forall x \in \mathbb{R}_+^n.$$

The proof relies on the weighted Arithmetic Mean-Geometric Mean (AM-GM) inequality, which states $M_{1,0}(Y, w) \geq 0$. Applying this to the terms $y_i = \frac{\alpha_i}{\beta_i} x_i^{2d}$ with weights $w_i = \alpha_i$ (where $\sum w_i = 2d$), we get:

$$\frac{\sum_{i=1}^n \beta_i x_i^{2d}}{2d} \geq (\prod_{i=1}^n (\frac{\alpha_i}{\beta_i})^{\alpha_i})^{1/2d} x^{2d}.$$

$$\Rightarrow \sum_{i=1}^n \beta_i x_i^{2d} \geq 2d(\prod_{i=1}^n (\frac{\alpha_i}{\beta_i})^{\alpha_i})^{1/2d} x^{\alpha} \quad (\dagger).$$

From our initial assumption $(*)$, we can take the $2d$-th root and rearrange to find:

$$\mu \leq 2d(\prod_{i=1}^n (\frac{\alpha_i}{\beta_i})^{\alpha_i})^{1/2d}.$$

Combining these results, we can chain the inequalities:

$$\mu x^{\alpha} \leq (2d(\prod_{i=1}^n (\frac{\beta_i}{\alpha_i})^{\alpha_i})^{1/2d}) x^{\alpha} \leq \sum_{i=1}^n \beta_i x_i^{2d}.$$

Thus, we have shown $\sum_{i=1}^n \beta_i x_i^{2d} \geq \mu x^{\alpha}$, which holds for $x_i > 0$. If any $x_i = 0$, then $x^{\alpha} = 0$ and the inequality becomes $\sum \beta_i x_i^{2d} \geq 0$, which is trivially true.

Part 2: ($\Rightarrow$)
Now, we prove the other direction by contradiction. Assume that $f(x)$ is PSD, but the inequality $(*)$ does **not** hold. This means:

$$\mu^{2d} \prod_{i=1}^n \alpha_i^{\alpha_i} > (2d)^{2d} \prod_{i=1}^n \beta_i^{\alpha_i}$$

Taking the $2d$-th root and rearranging gives:

$$\mu > 2d(\prod_{i=1}^n (\frac{\beta_i}{\alpha_i})^{\alpha_i})^{1/2d}.$$

Since we assumed $f(x) \geq 0$ for all $x \in \mathbb{R}_+^n$, we can show a contradiction by constructing a specific vector x for which $f(x) < 0$. We choose the vector x that corresponds to the equality condition of the AM-GM inequality used in *Part 1*. This occurs when all the terms $y_i = \frac{\alpha_i}{\beta_i} x_i^{2d}$ are equal. Let's choose x such that they are all equal to 1. Let $x_i = (\frac{\beta_i}{\alpha_i})^{1/2d}$.
Now we evaluate $f(x)$ for this specific x:

1. The sum term is:

$$\sum_{i=1}^n \beta_i x_i^{2d} = \sum_{i=1}^n \beta_i ((\frac{\beta_i}{\alpha_i})^{1/2d})^{2d} = \sum_{i=1}^n \beta_i (\frac{\beta_i}{\alpha_i}) = \sum_{i=1}^n \alpha_i = 2d.$$

2. The monomial term is:

$$x^\alpha = \prod_{i=1}^{n} x_i^{\alpha_i} = \prod_{i=1}^{n} ((\frac{\beta_i}{\alpha_i})^{1/2d})^{\alpha_i} = \prod_{i=1}^{n} (\frac{\beta_i}{\alpha_i})^{\alpha_i/2d} = (\prod_{i=1}^{n} (\frac{\beta_i}{\alpha_i})^{\alpha_i})^{1/2d}.$$

Substituting these into $f(x)$:

$$f(x) = 2d - \mu(\prod_{i=1}^{n} (\frac{\alpha_i}{\beta_i})^{\alpha_i})^{1/2d}.$$

This implies that $2d - \mu(\prod(\frac{\alpha_i}{\beta_i})^{\alpha_i})^{1/2d} < 0$. So, for our chosen x, we have found that $f(x) < 0$. This contradicts the premise that $f(x)$ is PSD. Therefore, our assumption that the inequality $(*)$ does not hold must be false.
Both directions of the proof are complete. □

Theorem 2.2
Let $\alpha(0), \alpha(1), \ldots, \alpha(r) \in \mathbb{N}^n$ be exponent vectors. Let the exponent vector $\beta \in \mathbb{N}^n$ be a convex combination of the $\alpha(j)$ vectors, such that: $\beta = \sum_{j=0}^{r} \lambda_j \alpha(j)$, with $\lambda_j > 0$ and $\sum_{j=0}^{r} \lambda_j = 1$.
Consider the polynomial $f(X)$ with real coefficients $f_\alpha > 0$ defined as:

$$f(X) = \sum_{j=0}^{r} f_{\alpha(j)} X^{\alpha(j)} - f_\beta X^\beta$$

where $X = (X_1, \ldots, X_n)$.
The polynomial $f(X)$ is positive semidefinite (PSD) for all $X \in \mathbb{R}_+^n$ if and only if:

$$|f_\beta| \leq \prod_{j=0}^{r} (\frac{f_{\alpha(j)}}{\lambda_j})^{\lambda_j}$$

Proof Sketch:
The proof of the ($\Leftarrow$) direction relies on the weighted Arithmetic Mean-Geometric Mean (AM-GM) inequality. Let us define a set of terms Y_j and use the weights λ_j:

$$Y_j = \frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)} \quad j = 0, \ldots, r$$

The weighted AM-GM inequality states that the arithmetic mean is greater than or equal to the geometric mean:

$$\sum_{j=0}^{r} \lambda_j Y_j \geq \prod_{j=0}^{r} Y_j^{\lambda_j}$$

1. **The Arithmetic Mean (LHS):**

$$\sum_{j=0}^{r} \lambda_j Y_j = \sum_{j=0}^{r} \lambda_j \frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)} = \sum_{j=0}^{r} f_{\alpha(j)} X^{\alpha(j)}$$

2. **The Geometric Mean (RHS):**

$$\prod_{j=0}^{r} Y_j^{\lambda_j} = \prod_{j=0}^{r}((\frac{f_{\alpha(j)}}{\lambda_j})X^{\alpha(j)})^{\lambda_j} = \prod_{j=0}^{r}(\frac{f_{\alpha(j)}}{\lambda_j})^{\lambda_j}\prod_{j=0}^{r}(X^{\alpha(j)})^{\lambda_j}$$

The second part of the product simplifies using the definition of β:

$$\prod_{j=0}^{r} X^{\lambda_j\alpha(j)} = X^{\sum_{j=0}^{r}\lambda_j\alpha(j)} = X^{\beta}$$

So, the geometric mean is $(\prod_{j=0}^{r}(\frac{f_{\alpha(j)}}{\lambda_j})^{\lambda_j})X^{\beta}$.
Combining these, the AM-GM inequality implies:

$$\sum_{j=0}^{r} f_{\alpha(j)}X^{\alpha(j)} \geq \prod_{j=0}^{r}(\frac{f_{\alpha(j)}}{\lambda_j})^{\lambda_j})X^{\beta}$$

If we assume the theorem's condition holds ($f_\beta \leq \prod(\frac{f_{\alpha(j)}}{\lambda_j})^{\lambda_j}$), then it follows that:
$\sum_{j=0}^{r} f_{\alpha(j)}X^{\alpha(j)} \geq f_\beta X^\beta \Rightarrow f(X) \geq 0$.

This proves one direction of the theorem. □

Application to D-T forms

Problem

Find the condition under which the function $f(x)$ is positive semidefinite for $x \in \mathbb{R}_+^n$:

$$f(x) = \sum_{i=1}^{n} \beta_i x_i^{2d} - \mu x^\alpha$$

where $\beta_i, \mu > 0$ and the exponents satisfy $\sum_{i=1}^{n} \alpha_i = 2d$.

Solution via Theorem 2.2

We can map this problem to the general theorem's framework.

- The "arithmetic mean" part is $\sum_{i=1}^{n} \beta_i x_i^{2d}$. The corresponding exponent vectors are $\alpha(i) = 2de_i$, where e_i is the i-th standard basis vector in $\mathbb{R}^n$. The coefficients are $f_{\alpha(i)} = \beta_i$.
- The "geometric mean" part is μx^α. The exponent vector is $\beta = (\alpha_1, \alpha_2, \ldots, \alpha_n)$. The coefficient is $f_\beta = \mu$.
- We must express β as a convex combination of the $\alpha(i)$ vectors: $\beta = \sum_{i=1}^{n} \lambda_i \alpha(i)$.

$$(\alpha_1, \ldots, \alpha_n) = \sum_{i=1}^{n} \lambda_i(0, \ldots, 2d, \ldots, 0) = (2d\lambda_1, \ldots, 2d\lambda_n)$$

This implies that the weights must be $\lambda_i = \frac{\alpha_i}{2d}$.

- The condition $\sum \lambda_i = 1$ becomes $\sum \frac{\alpha_i}{2d} = 1$, which is exactly the given condition $\sum \alpha_i = 2d$.

Now, we apply the inequality from the general theorem:

$$f_\beta \leq \prod_{i=1}^{n} (\frac{f_{\alpha(i)}}{\lambda_i})^{\lambda_i}.$$

Substituting our specific terms:

$$\mu \leq \prod_{i=1}^{n} (\frac{\beta_i}{\alpha_i/2d})^{\alpha_i/2d} = \prod_{i=1}^{n} (\frac{2d \cdot \beta_i}{\alpha_i})^{\alpha_i/2d}$$

To see that this is the same condition, we can raise both sides to the power of $2d$:

$$\mu^{2d} \leq (\prod_{i=1}^{n} (\frac{2d \cdot \beta_i}{\alpha_i})^{\alpha_i/2d})^{2d} = \prod_{i=1}^{n} (\frac{2d \cdot \beta_i}{\alpha_i})^{\alpha_i},$$

Since $\sum \alpha_i = 2d$, this simplifies to:

$$\mu^{2d} \leq \frac{(2d)^{2d} \prod \beta_i^{\alpha_i}}{\prod \alpha_i^{\alpha_i}}.$$

Thus

$$\mu^{2d} \prod \alpha_i^{\alpha_i} \leq (2d)^{2d} \prod \beta_i^{\alpha_i}. \square$$

Section 4:

In this section we investigate $M_{2d,p}(\underline{X}, \alpha)$ where $|\alpha| = 2d$, and $p|2d$, for values $d = 1, 2, 3$.

Proposition 4.1 $M_{2d,0}(X, \alpha)$ is psd D-T, SOBS.
Proof. See [GM01] *and see if the proof can be modified in a simpler format.*
MISSING: Add the proof for SOBS.

Theorem 4.1 $M_{2d,p}(X, \alpha)$ with $d = 1, 2, 3$, $p|2d$ is SOS for $\forall n \in \mathbb{N}$ and $\alpha \in \mathbb{N}^n$ with $|\alpha| = 2d$.
Proof:

$d = 1$:

$M_{2,0}, M_{2,1}$ are psd quadratic forms and by Hilbert's theorem, they are SOS.

$d = 2$:

For the case $p = 0$, we already know they are psd and even SOBS bur Pop 4.1.
For the case $p = 1$, we show that $M_{4,1}(X, \alpha)$ is a Sum of Squares.
We provide a detailed proof for the proposition that the family of nonnegative quartic forms $M_{4,1}$ is always a SOS.

In other words, for any number of variables n and any nonnegative weights $\alpha = (\alpha_1, \ldots, \alpha_n)$ such that $|\alpha| = \sum_{i=1}^{n} \alpha_i = 4$, the form

$$M_{4,1}(X, \alpha) = \frac{\sum_{i=1}^{n} \alpha_i X_i^4}{4} - (\frac{\sum_{i=1}^{n} \alpha_i X_i}{4})^4$$

is a SOS.
For the proof we examine each case by the number of variables n.

1. **Case n=2 (Binary Forms):** For $n = 2$, the form $M_{4,1}$ is a PSD binary quartic form. By **Hilbert's Theorem (1888)**, any nonnegative polynomial in two variables is a Sum of Squares. Therefore, $M_{4,1}$ is SOS for $n = 2$.
2. **Case n=3 (Ternary Quartics):** For $n = 3$, the form $M_{4,1}$ is a PSD ternary quartic. Again, by **Hilbert's Theorem**, any nonnegative ternary quartic is a Sum of Squares. Therefore, $M_{4,1}$ is SOS for $n = 3$.
3. **Case n=4:**
   - First, we assume uniform weights, $\alpha = (1, 1, 1, 1)$. The form is:

     $$M_{4,1}(X, (1, 1, 1, 1)) = \frac{\sum_{i=1}^{4} X_i^4}{4} - (\frac{\sum_{i=1}^{4} X_i}{4})^4$$

     It is known that this specific form is a Sum of Squares. This has been verified computationally by running a **Semidefinite Program (SDP)**.
     -[ ]**TODO:** Find an explicit SOS decomposition for this form.
   - For the remaining weight vectors α where $|\alpha| = 4$ (e.g., $\alpha = (2, 1, 1, 0)$, $\alpha = (3, 1, 0, 0)$, $\alpha = (4, 0, 0, 0)$), all of them involve at most three non-zero entries. This means they are effectively ternary quartics (or simpler). We can therefore reduce to the $n = 3$ case and conclude they are SOS.
4. **Case $n \geq 5$:**
   Using the constraint $|\alpha| = \sum_{i=1}^{n} \alpha_i = 4$, we see that for any vector $\alpha = (\alpha_1, \ldots, \alpha_n)$, we will have at most four non-zero entries α_i. All other entries must be zero.
   This means the polynomial $M_{4,1}(X, \alpha)$ will only involve variables corresponding to the non-zero weights. Therefore, we can reduce any case with $n \geq 5$ variables to a case with at most 4 variables $(n \leq 4)$.
   Since all cases for $n \leq 4$ are shown to be SOS, and any case with $n \geq 5$ reduces to one of these, the proof is complete for the case $p = 1$.
   For the case $p = 2$, we show that $M_{4,2}(X, \alpha)$ is SOS.
   This section provides a proof for the proposition that the family of nonnegative quartic forms $M_{4,2}$ is also always a Sum of Squares.
   For any number of variables n and any nonnegative weights $\alpha = (\alpha_1 \ldots, \alpha_n)$ such that $|\alpha| = \sum_{i=1}^{n} \alpha_i = 4$, the form

$$M_{4,2}(X,\alpha) = \frac{\sum_{i=1}^{n} \alpha_i X_i^4}{4} - (\frac{\sum_{i=1}^{n} \alpha_i X_i^2}{4})^2$$

is SOS.
For the proof we examine each case by the number of variables n.

5. **Case $n = 2$ (Binary Forms):** For $n = 2$, the form $M_{4,2}$ is a PSD binary form. By **Hilbert's Theorem**, it is therefore a Sum of Squares.
6. **Case $n = 3$ (Ternary Quartics):** For $n = 3$, the form $M_{4,2}$ is a PSD ternary quartic. By **Hilbert's Theorem**, it is therefore a Sum of Squares.
7. **Case $n = 4$:**
   - First, we assume uniform weights, $\alpha = (1,1,1,1)$. The form is:

   $$M_{4,2}(X,(1,1,1,1)) = \frac{\sum_{i=1}^{4} X_i^4}{4} - (\frac{\sum_{i=1}^{4} X_i^2}{4})^2$$

   We know this form is nonnegative, computation shows that $f_{SOS} = 0$ and hence it is SOS.
   -[ ] **TODO:** Find an explicit SOS decomposition for this form.
   For the remaining weight vectors α where $|\alpha| = 4$ (e.g., $\alpha = (2,1,1,0)$, $\alpha = (3,1,0,0)$, $\alpha = (4,0,0,0)$), all of them involve at most three non-zero entries. This means we can reduce to the case $n < 4$.
8. **Case $n \geq 5$:**
   Using the constraint $|\alpha| = \sum_1^n \alpha_i = 4$, it is the same argument as before. Any vector α will have at most four non-zero entries.
   This reduces any problem with $n \geq 5$ variables to an equivalent problem with at most 4 variables ($n \leq 4$).
   Since all cases are either directly covered by Hilbert's Theorem or reduce to a previous case, the proof is complete for $p = 2$.

$d = 3$

The next proof is for the family of forms:

$$M_{6,p}((X_1,\ldots,X_n),(\alpha_1,\ldots,\alpha_n))$$

where the parameters p and the weights α satisfy:

- $p = 0$: $M_{6,0}$,
- $p = 1$: $M_{6,1}$,
- $p = 2$: $M_{6,2}$,
- $p = 3$: $M_{6,3}$.
  We prove by examining the number of variables n of the family of forms $M_{q,p}(X,\alpha)$ for degree $q = 6$. The parameters under consideration are:
- $q = 6$
- $p \in \{0,1,2,3\}$

- The sum of weights $|\alpha| = \sum \alpha_i = 6$.

1. **Case $n = 2$ (Binary Forms):** For $n = 2$, we are in the binary case. By **Hilbert's Theorem**, any nonnegative binary form is a Sum of Squares. Therefore, all forms $M_{6,p}$ are SOS for $n = 2$.
2. **Case $n = 3$ (Ternary Forms):**
   - For $p = 0$, $M_{6,0}$ is known to be SOS.
   - For $p = 1$, we need to analyze the form:

     $$M_{6,1}(X, \alpha) = \frac{\sum_{i=1}^{3} \alpha_i X_i^6}{6} - (\frac{\sum_{i=1}^{3} \alpha_i X_i}{6})^6$$

     The analysis then proceeds by examining the possible integer partitions of 6 into 3 parts for the weight vector $\alpha \in \mathbb{N}^3$. The possible partitions are:

     $$\alpha \in \{(2,2,2), (3,2,1), (4,1,1), (5,1,0), (6,0,0)\}$$

     - **For $\alpha = (2,2,2)$:** The form is $M_{6,1}(X, (2,2,2))$. It is stated that in this case, the form is known to be SOS.
       -[ ] **TODO:** Find an explicit SOS decomposition.
     - **For $\alpha = (3,2,1)$:** The form is $M_{6,1}(X, (3,2,1))$. Again, it is stated that this form is SOS.
     - **For $\alpha = (4,1,1)$:** The same goes for this case; the form is SOS.
     - The cases for $\alpha = (3,2,1)$ and $\alpha = (4,1,1)$ can be reduced to the case of $\alpha = (2,2,2)$ by an appropriate change of variables, which proves their SOS property.

Case $n = 4$

We consider the form $M_{6,1}(X, \alpha)$ for $n = 4$ variables, with the constraint $|\alpha| = 6$. The possible integer partitions of 6 into 4 parts for the weight vector α are:

$$\alpha \in \{(2,2,1,1), (3,1,1,1), (4,1,1,0), (4,2,0,0), (5,1,0,0), (3,2,1,0), (6,0,0,0)\}$$

The cases including at least one 0, e.g., $(4,1,1,0), (4,2,0,0), (5,1,0,0), (3,2,1,0), (6,0,0,0)$, all are effectively forms in fewer than 4 variables ($n \leq 3$), which have already been handled. Thus, these cases reduce to the $n \leq 3$ analysis.
The analysis then focuses on the remaining cases:

- **For $\alpha = (2,2,1,1)$:** The form $M_{6,1}(X, (2,2,1,1))$ is stated to be SOS because it is a known nonnegative form and SDP decomposition shows it is SOS.
- **For $\alpha = (3,1,1,1)$:** The same argument applies; the form is SOS.

Case $n = 5$ and $n = 6$

- **For $n = 5$:** The partitions of 6 into 5 parts are listed, with cases like $(2, 1, 1, 1, 1)$ and $(3, 1, 1, 1, 0)$ highlighted. Again, any partition with a zero entry reduces to a case with fewer variables.
- **For $n = 6$:** The only partition without zeros is the uniform case, $\alpha = (1, 1, 1, 1, 1, 1)$. All others reduce.
  This completes the proof for Theorem 4.1. □

Remark 4.1:

1. We would like to investigate if Theorem 4.1 generalizes to any d.
2. It would be interesting to know if these forms are SONC cone.

Section 5: Beyond the SOSONC cone

Section 5.1: Specific Examples to Analyze:

- $Q(X, Y, Z, W) = X^4 + Y^4 + Z^4 + W^4 - 4XYZW$.
  Similar as the Motzkin form, one can easily verify that both Choi Lam forms Q,S, and the modified Motzkin M are indeed PSD circuits (page 31, M. Shick thesis).

  $$R(X, Y, Z) := X^6 + Y^6 + Z^6 - (X^4Y^2 + X^4Z^2 + X^2Y^4 + X^2Z^4 + Y^4Z^2 + Y^2Z^4) + 3X^2Y^2Z^2$$

  $$\hat{R}(X, Y, Z, W) := X^2(X - W)^2 + Y^2(Y - W)^2 + Z^2(Z - W)^2 + 2XYZ(X + Y + Z - 2W)$$

  $$P(X, Y, Z) := 200[(X^3 - 4XZ^2)^2 + (Y^3 - 4YZ^2)^2] + (Y^2 - X^2)X(X + 2Z)[X(X - 2Z) + 2(Y^2 -$$

  Remarkably, there is the following relation between Q and R (see [Rez07, p. 5]):

  $$\hat{R}(X - W, Y - W, Z - W, X + Y + Z - W) = 2Q(X, Y, Z, W).$$

  Since the PSD and SOS cones are closed under linear transformations, the identity above shows that R is PSD (SOS) if and only if Q is PSD (SOS). Now, using the Newton polytope method one can prove that $Q \in P^h_{4,4} \setminus \Sigma^h_{4,4}$. In this sense, the Robinson form R admits a proof of $\hat{R} \in P^h_{4,4} \setminus \Sigma^h_{4,4}$ following the Newton polytope method. (page 38)

Section 5.2: Separating forms via linear transformation of the variables

Section 6: Positivestellensatz and relation to the cone of $2d$ powers

Section 7: Application to polynomial optimization.

Acknowledgement

The research was started during a research in teams stay in *Banff International Research Station* (BIRS), September 21st-October 04, 2025. We wish to thanks BIRS for their hospitality.

References:

- [JBL01] J. B. Lasserre. *Global optimization with polynomials and the problem of moments.* SIAM J. Optim., 11(3):796–817, 2001.
- [GM01] M. Ghasemi and M. Marshall, *Lower bounds for polynomials using geometric programming*, SIAM J. Optim. 22 (2012), no. 2, 460–473.
- [GLM] M. Ghasemi, J.B. Lasserre, and M. Marshall, *Lower bounds on the global minimum of a polynomial*, Comput. Optim. Appl. 57 (2014), no. 2, 387–402.
- [GM02]M. Ghasemi and M. Marshall, *Lower bounds for a polynomial on a basic closed semialgebraic set using geometric programming*, 2013, Preprint, arxiv:1311.3726.
- [GM03] M. Ghasemi, M. Marshall, *Lower Bound for a Polynomial on a product of hyperellipsoids using geometric programming*, Update the arXiv reference.
- [DIdW01] M. Dressler, S. Iliman, and T. de Wolff, *An Approach to Constrained Polynomial Optimization via Nonnegative Circuit Polynomials and Geometric Programming*, To appear in the Journal of Symbolic Computation (MEGA 2017 special issue)
- [DIdW02] M. Dressler, S. Iliman, and T. de Wolff. *A Positivstellensatz for Sums of Nonnegative Circuit Polynomials*. SIAM J. Appl. Algebra Geom., 1(1):536–555, 2017.
- [IdW] Sadik Iliman and Timo De Wolff, *Lower Bounds for Polynomials with Simplex Newton Polytopes Based on Geometric Programming,* SIAM Journal on Optimization 26, no. 2 (January 2016): 1128–46, https://doi.org/10.1137/140962425;
- [MSc01] Moritz Schick *Sums of Squares plus Sums of Nonnegative Circuit Polynomials*, PhD. Dissertation. 2025.
- [Rez07] B. Reznick. *On Hilbert's construction of positive polynomials*, 2007. Preprint, arXiv:0707.2156.
- [] Mareike Dressler und Moritz Schick, *Study of the cone of sums of squares plus sums of nonnegative circuit forms* Advances in Geometry, 25(1), 127-146 (2025)

Mehdi Ghasemi: mehdi.ghasemi@usask.ca
Edmonton Police Service

Salma Kuhlmann: salma.kuhlmann@uni-konstanz.de
University of Konstanz